Scalable Machine Learning 2026 Assignment

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COM6012 Scalable Machine Learning

The University of Sheffield

Introduction

This report presents a brief summary of the results obtained from the assignment, together with explanations of the tasks, selected figures, and code snippets. It describes the workflow used to run each question, the outputs and figures generated, and the main observations from the results.

To run each question make sure that the file path to dataset is correct and run code by using "sbatch Q$_script.sh".

Q1

Section A: Counts by Categories

The table below shows the total number of requests and the number of unique hostnames for each UK sector category in the NASA dataset.

Sector	Total Requests	Unique Hosts
UK academic institutions(ac.uk)	25012	1023
UK companies (co.uk)	22503	1118
UK government departments (gov.uk)	195	15

Table 1: Total statistics of requests

Section B: Academic Hosts

Institution domain Extraction and Aggregation

To extract the institution domain, each academic hostname is split into its components using the dot separator. The last three components are then selected and joined back together. For example, a hostname such as ***www.shef.ac.uk*** becomes ***shef.ac.uk***.

```
academicInstitutions = academicHosts.select(
    expr("concat_ws('.', slice(split(host, '\\\\.'), -3, 3)) as institution")
)
```

```
institutionCounts = academicInstitutions.groupBy("institution").count()
```

Code Snippet 1: institution domain extraction

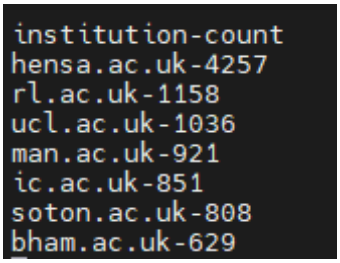
After extracting the institution domain, the data is aggregated by grouping rows with the same institution and counting the number of requests for each unique group. This produces the sorted total number of requests made to each UK academic institution.

```
institutionCounts.orderBy(col("count").desc()).show()
```

Code Snippet 2: Sorting method for table

CSV output directory/file

The code produces a table in the output and saves the CSV result in the ***Q1_table1.csv***. This CSV file contains the aggregated request counts for each academic institution domain, representing each institution together with its number of requests. In addition, the output file explicitly reports that the University of Sheffield domain, ***shef.ac.uk***, recorded 623 requests, and that 7 other UK academic institutions had more requests than Sheffield. The CSV output file is shown below.

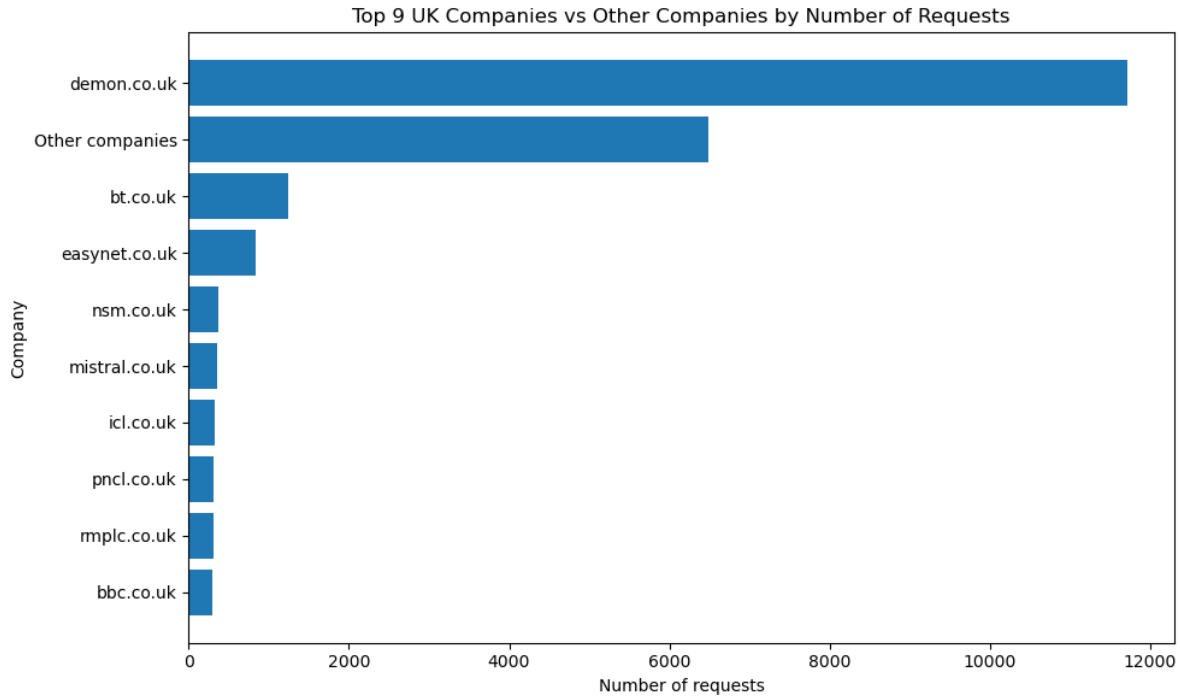


Q1_table1.csv output file

Section C: Company Hosts

A horizontal bar chart was chosen due to it provides a clear comparison of request counts across company domains. This type of chart is suitable for categorical data and makes it easier to compare the top 9 UK company domains with the total of all remaining companies.

A horizontal style was used because the company domain labels are text-based and longer names are easier to read on the y-axis. The chart below also highlights the strong dominance of demon.co.uk compared with the other company domains.



Q1_BarChart: comparing the number of requests for the top 9 UK company domains with all other company domains combined.

Section D: Heatmaps

Academic_heatmap figure shows the access pattern for the most active UK academic institution domain. Request activity is concentrated in specific daytime periods, with several visible peaks across the month.

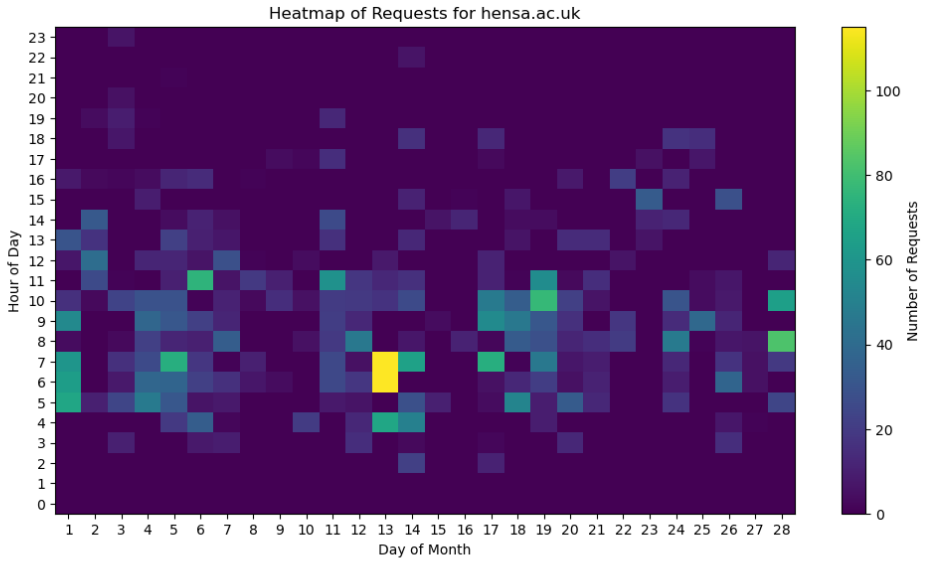


Figure: Heatmap showing request intensity by day of month and hour of day for the most active UK academic institution domain (hensa.ac.uk). Institution level is defined using the last three labels of the hostname.

Q1_sectionD_academic_heatmap: The most active UK academic institution domain heatmap (hensa.ac.uk)

Company_heatmap shows the access pattern for the most active UK company domain.

Activity is spread across more days and hours, with several strong peaks in request volume.



Figure: Heatmap showing request intensity by day of month and hour of day for the most active UK company domain (demon.co.uk). Company domain level is defined using the last three labels of the hostname.

Q1_sectionD_company_heatmap: demon.co.uk heatmap (demon.co.uk)

Government_heatmap shows the combined access pattern for all UK government domains. Compared with the other two heatmaps, request activity is much lower and more sparse.

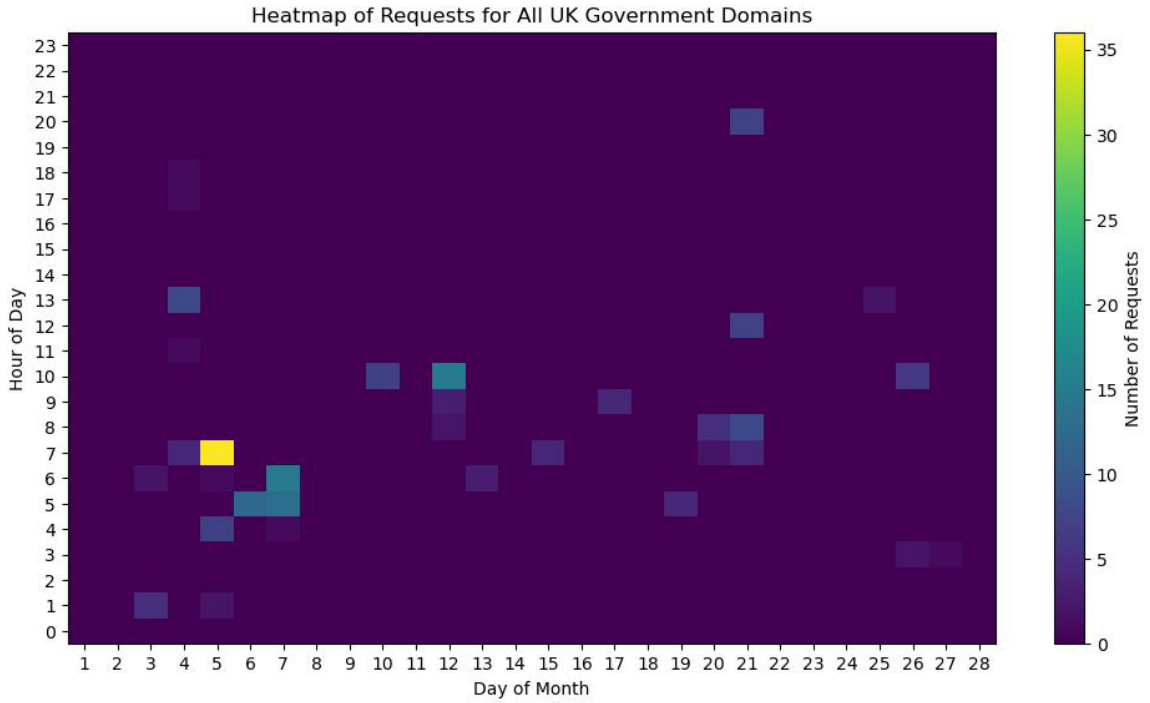


Figure: Heatmap showing the number of requests by day of month and hour of day for all UK government hosts ending with .gov.uk combined.

Q1_sectionD_government_heatmap: government combined heatmap

Section E: Observations

Observation 1

The demon.co.uk created about 11,719 out of 22,503 total UK company requests (approximately 52%), far exceeding even the combined "Other companies" group of 6,490 requests. This is likely because Demon Internet was one of the UK's largest consumers in 1995, meaning a large share of individual UK users accessed the web through its infrastructure. This suggests the UK commercial internet in 1995 was highly centralised around a few dominant providers, and NASA could leverage this by prioritising content

optimisation or mirror servers targeting high-volume domains to improve access performance for UK users.

Observation 2

UK government domains recorded only 195 total requests from 15 unique hostnames in July 1995, which is dramatically lower than academic (25,012) and company (22,503) traffic, with the government heatmap showing a highly irregular access pattern lacking any consistent pattern. This is likely because UK government internet adoption in 1995 was still in its very early stages, with most departments having limited or no direct internet connectivity compared to academic institutions and commercial domains. This suggests that the UK government sector was largely disconnected from the emerging public internet at the time, and NASA could use this insight to prioritise outreach and content delivery resources toward academic and commercial networks where engagement was substantially higher.

Q2

Section A: Preprocessing

After applying the preprocessing steps (extracting month and weekday, uppercasing direction_of_travel, dropping NULL rows, one-hot encoding categorical features, and standardising latitude and longitude), the dataset was split by year for sets(see Table 2)

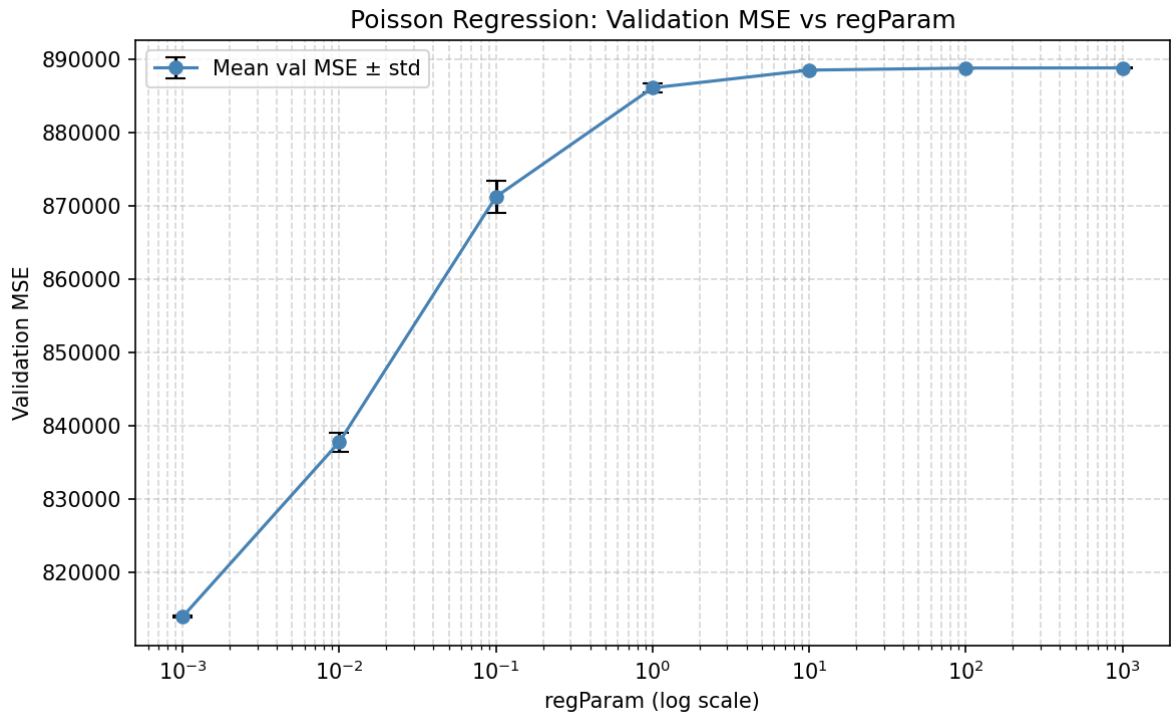
Set	Year Range	Sample Size
Training	2000-2021	4,655,073
Validation	2022-2023	306,804
Testing	2024	151,848
Total	2000-2024	5,113,725

Table 2

Section B: Poisson regression

Student Registration Number: 250117976

The validation curve shows the mean validation MSE across five random training subsets for each value of regParam, with error bars representing ± one standard deviation. The lowest mean validation MSE was obtained at regParam = 0.001 (approximately 813,000), and the MSE increased as regParam grew larger, stabilizng around 888,000 for regParam ≥ 10. The error bars are very small across all values, indicating that the results are stable and consistent across the five different random subsets. This suggests that the Poisson regression model benefits from minimal regularisation on this dataset, as stronger penalties narrows the coefficients rapidly and reduce predictive performance. Therefore, the optimal value selected for retraining on the combined training and validation sets was regParam = 0.001.



Q2_B_validation_curve.png

Test set and coefficients

Full txt file of final result for the model on the test set is ***Q2_B_final_results.txt***

Optimal regParam: 0.001

Test MSE: 787575.1086

Intercept: 6.270922303416401

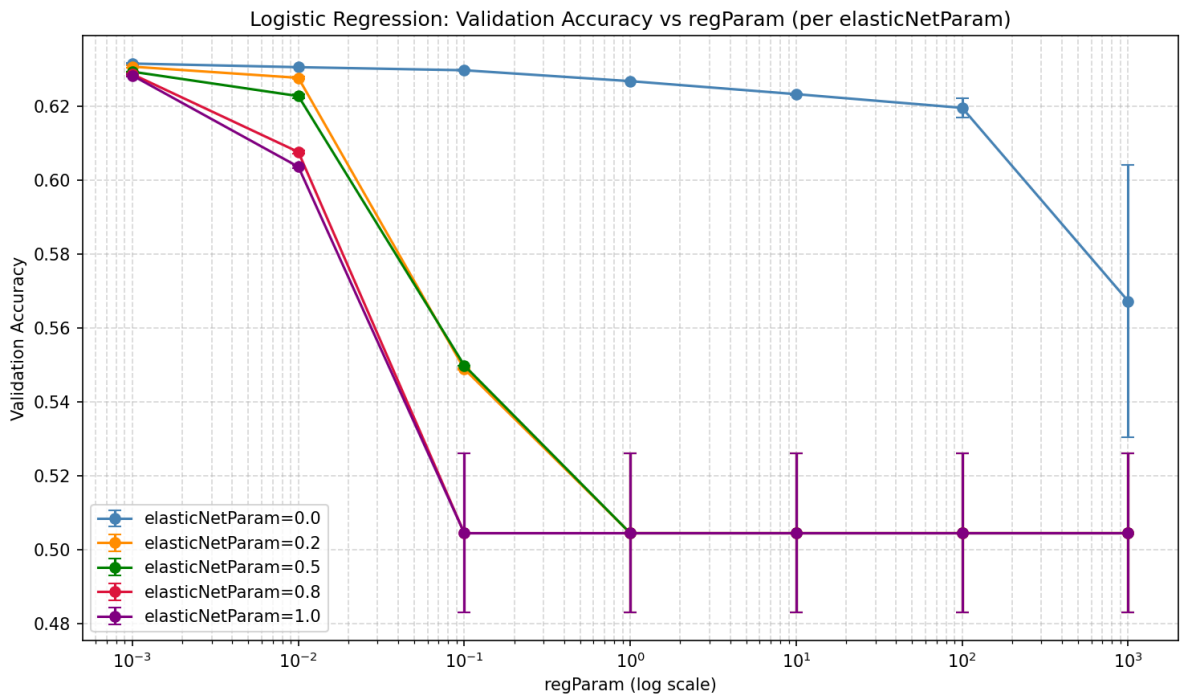
Number of coefficients: 266

Coefficients file: Q2_B_coefficients

Section C: Logistic Regression

Reg number: 250117976

The validation curve shows that the highest mean validation accuracy was achieved with regParam = 0.001 and elasticNetParam = 0.0, giving an accuracy of approximately **0.632** across the five random training subsets. This suggests that the logistic regression model benefits from only minimal regularisation, since increasing regParam generally reduced validation accuracy and led to underfitting, specifically for larger values where accuracy dropped close to **0.504**. The best performance with elasticNetParam = 0.0 indicates that L2 regularisation was more suitable than L1 or mixed elastic-net regularisation, likely because the one-hot encoded categorical features contain useful information that should be shrunk rather than removed. The small error bars also show that the results were stable across the five runs. Therefore, the optimal hyperparameters selected for the final model were regParam = 0.001 and elasticNetParam = 0.0.



Q2_C_validation_curve.png

Test set and coefficients

Optimal regParam: 0.001

Optimal elasticNetParam: 0.0

Test Accuracy: 0.6412

Intercept: -0.003436812053028364

Number of coefficients: 266

Coefficients: Q2_C_coefficients

Observation

Observation from the results is that both the Poisson and logistic regression models achieved their best performance under the weakest regularisation setting. For the Poisson regression model, the lowest validation MSE occurred at regParam = 0.001, after which the error increased as the regularisation became larger. A similar pattern

was observed for logistic regression, where the highest validation accuracy was obtained with regParam = 0.001 and elasticNetParam = 0.0. This indicates that L2 regularisation with minimal penalty was most appropriate for the processed feature set. Stronger regularisation, particularly when combined with higher elastic-net values, likely reduced model flexibility by removing informative coefficients from the one-hot encoded categorical and spatial features. As a result, the models showed signs of underfitting at larger regularisation values. The final logistic regression model achieved a test accuracy of **0.6412**, suggesting that the weakly regularised model was able to generalise reasonably well.

Q3

All results can be verified from Q3_output.txt

Section A: 2% sample

Best Parameters

The 2% stratified sample contained 220,163 instances and preserved a balanced class distribution. Random Forest and GBT models were tuned using Spark pipelines, parameter grids, and 3-fold cross-validation, with three hyperparameters optimised for each model. The best configurations are shown below.

RF numTrees	150
RF maxDepth	12
minInstancesPerNode	5

Table 3: Random Forest Cross-Validation best Parameters

GBT maxIter	30
GBT maxDepth	5
GBT stepSize	0.2

Table 4: GBT Cross-Validation Best Parameters

Train and Test set Results with best parameters

Random Forest achieved higher training accuracy than GBT (76.68% vs 72.61%), but GBT obtained slightly higher test accuracy (71.92% vs 71.67**%**). This means that Random Forest fitted the training subset more strongly, while GBT generalised marginally better. Overall, the difference in test performance was small, indicating comparable performance on the 2% sample.

	Train Accuracy	Test Accuracy
Random Forest	76.68%	71.67%
GBT	72.61%	71.92%

Table 5: RF and GBT results on 2% sample

Section B: Full Dataset

Set size

The full dataset was split into 8,800,885 training instances and 2,199,115 test instances.

Train size	8800885
Test size	2199115

Table 6: Train and Test set size

Performance

On the full dataset, the GBT model obtained slightly higher accuracy on both the training set (72.02%) and the test set (71.95**%**) compared with the Random Forest model, which achieved 71.98**%** and 71.76**%**, respectively. The close agreement between training and test accuracy indicates that both models generalised well and did not show clear evidence of overfitting. Overall, GBT produced the stronger result on the full dataset, although the improvement over Random Forest was marginal.

	Train Accuracy	Test Accuracy
Random Forest	71.98%	71.76%
GBT	72.02%	71.95%

Table 6: RF and GBT results on full Dataset

Q4

Section A: Time-Split Recommendations

Settings

For each split, two ALS configurations were evaluated. Setting 1 used the default ALS configuration from Lab 6 (rank=10, maxIter=10, regParam=0.1. Setting 2 used rank=20, maxIter=15, and regParam=0.1. The motivation for Setting 2 is that with 20 million ratings, a rank of 10 may underfit the diversity of user and movie latent structure; doubling the rank allows ALS to capture finer-grained latent factors, while increasing maxIter to 15 provides the larger model with more iterations to converge. The regularisation parameter was kept at the default 0.1 so that any change in performance can be attributed to model capacity rather than regularisation strength.

	Rank	maxIter	regParam	Seed
Setting 1	10	10	0.1	250117976
Setting 2	20	15	0.1	250117976

Table 7: Setting parameters

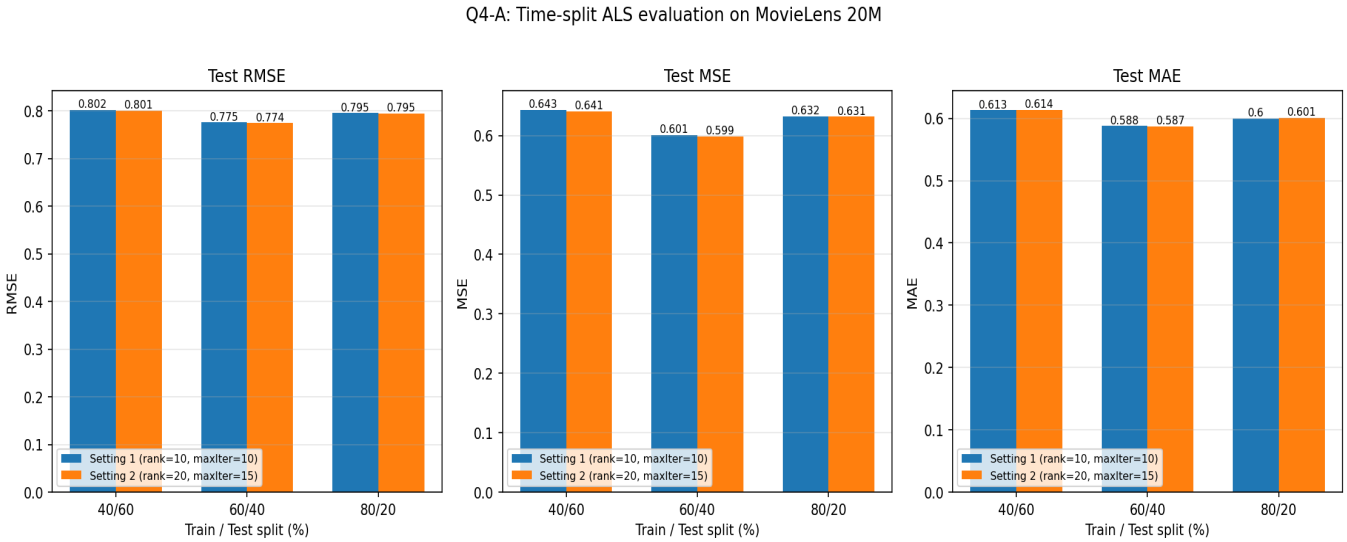
Metrics

For each of the three time-based splits and both ALS settings, the test set was evaluated using Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and Mean Absolute Error (MAE). The resulting 18 values are reported in Table 8 and visualised in Figure X. Setting 2 yields consistently lower RMSE and MSE than Setting 1 across all splits, suggesting a modest improvement attributable to the increased rank and iteration count, whereas MAE values remain essentially unchanged between the two configurations.

Train/Test Split	ALS setting	RMSE	MSE	MAE
40/60	Setting 1	0.8021	0.6433	0.6133
40/60	Setting 2	0.8006	0.6409	0.6138
60/40	Setting 1	0.7753	0.6011	0.5881
60/40	Setting 2	0.7738	0.5987	0.5871

80/20	Setting 1	0.7951	0.6321	0.6001
80/20	Setting 2	0.7947	0.6315	0.601

Table 8: ALS performance across time-based train/test splits



Q4_A_metrics.png

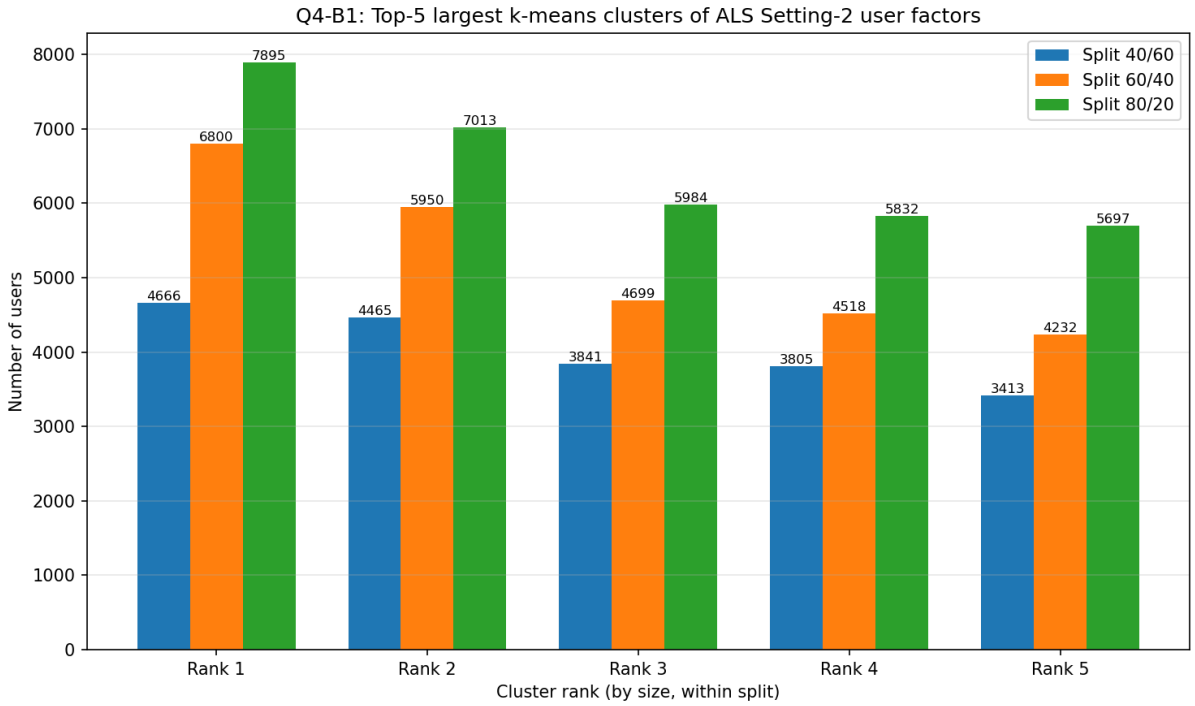
Section B:

User Analysis

For each of the three time-splits, k-means clustering with k = 25 was applied to the user factors learned by ALS Setting 2, using the last five digits of the student registration number (17976) as the random seed. The five largest user clusters were identified for each split, and their sizes are reported in Table 9 and visualised in figure below.

Rank	Split 40/60	Split 60/40	Split 80/20
1	4,666	6,800	7,895
2	4,465	5,950	7,013
3	3,841	4,699	5,984
4	3,805	4,518	5,832
5	3,413	4,232	5,697

Table 9: Top 5 cluster sizes per split



Q4_B_cluster_sizes

Genre Analysis

For each split, only the largest user cluster from Q4-B1 was considered. Within the training set, all movies rated by users in this cluster were collected together with their cluster-level average ratings (movies_largest_cluster). Movies with an average rating ≥ 4.0 were retained as top_movies, and their genres (pipe-separated in movies.csv) were exploded and counted. The ten most popular genres for each split are reported in Table 10.

Rank	Split 40/60	Split 60/40	Split 80/20
1	Drama (561)	Drama (809)	Drama (1356)
2	Comedy (217)	Comedy (308)	Comedy (536)
3	Romance (163)	Romance (222)	Romance (399)
4	Thriller (128)	Thriller (167)	Thriller (244)
5	Crime (108)	Crime (154)	Documentary (227)
6	Adventure (76)	Documentary (138)	Crime (220)
7	War (70)	War (112)	War (166)
8	Action (70)	Adventure (110)	Adventure (159)
9	Documentary (69)	Action (103)	Action (153)
10	Mystery (60)	Mystery (76)	Fantasy (119)

Table 10 : Top-10 most popular genres amog top_movies (rating >=4.0)

Section C : Observations

Observation 1

Across both ALS settings, the 60/40 split produced the lowest RMSE (0.7753 and 0.7738) and lowest MAE (0.5881 and 0.5871), outperforming both the 40/60 split and, more surprisingly, the 80/20 split . A possible cause is that in a chronological time-split, the 80/20 partition pushes the test set into the most recent rating period, which may contain newer users, newer movies, and shifted rating distributions that the model has had less exposure to during training, while the 60/40 split tests on a more representative middle period of the dataset. For a movie website such as Netflix, this observation suggests that simply maximising training data is not always optimal under realistic time-based evaluation, and that recommendation models should be evaluated on multiple times to account for concept drift in user preferences and content catalogues.

Observation 2

In every split, Drama is the most popular genre among ‘top_movies’ by a wide margin appearing 561, 809, and 1356 times for the 40/60, 60/40, and 80/20 splits respectively, more than twice as often as the second-ranked Comedy in each case. A possible cause is that Drama is an extremely broad genre label assigned to a large proportion of critically trend films, meaning that high-rated films across many sub-categories tend to be tagged as Drama by default in the MovieLens taxonomy, which influenses its count relative to more specific genres such as Mystery or Fantasy. For a movie website such as Netflix, this observation indicates that broad genre labels may be of limited use for fine grained personalisation, and

that combining genre information with finer grained tags (e.g. tag-genome scores) or sub-genres would more effectively differentiate user preferences within large clusters.